arXiv:2412.09565v2 [cs.LG] 8 Feb 2025

Obfuscated Activations Bypass LLM Latent-Space Defenses

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Abstract

Recent *latent-space* monitoring techniques have shown promise as defenses against LLM attacks. These defenses act as scanners that seek to detect harmful activations before they lead to undesirable actions. This prompts the question: Can models execute harmful behavior *via inconspicuous latent states*? Here, we study such *obfuscated activations*. We show that state-of-the-art latent-space defenses—including sparse autoencoders, representation probing, and latent OOD detection—are all vulnerable to obfuscated activations. For example, against probes trained to classify harmfulness, our attacks can often reduce recall from 100% to 0% while retaining a 90% jailbreaking rate. However, obfuscation has limits: we find that on a complex task (writing SQL code), obfuscation reduces model performance. Together, our results demonstrate that neural activations are highly malleable: we can reshape activation patterns in a variety of ways, often while preserving a network's behavior. This poses a fundamental challenge to latent-space defenses.[‡]

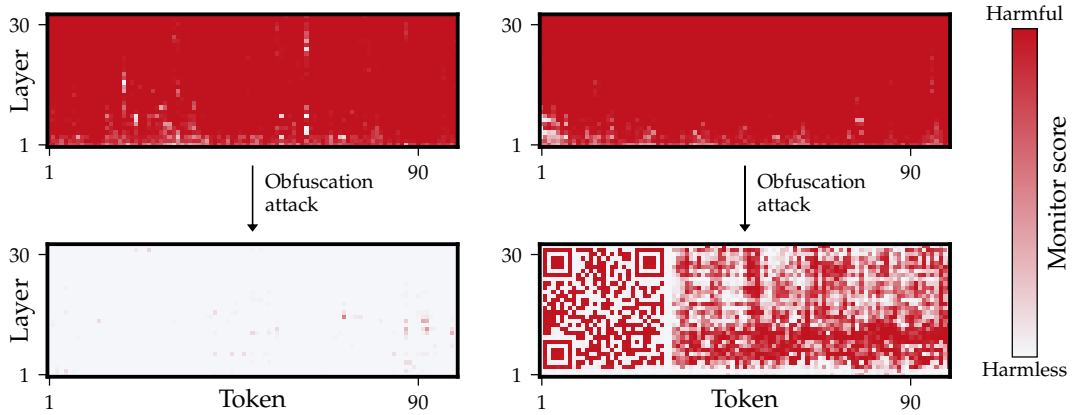

Figure 0: **Obfuscation attacks achieve a high degree of control over how a harmfulness monitor classifies activations *while also controlling model outputs*.** Each heatmap shows a harmfulness monitor scan of LLM latent-space activations during a harmful model output (instructions on how to smuggle a bomb past airport security). The sequence dimension is shown on the x-axis and the residual stream layers on the y-axis. Top: No obfuscation attack, harmfulness is clearly detected. Bottom: our obfuscation embedding attacks that bypass the monitor while inducing harmful generations. Left, we optimize the attack to bypass the monitor everywhere. Right, we demonstrate the power of obfuscated activation attacks by forcing the monitor to output a QR code containing a URL to our paper website.

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‡ Code available at https://github.com/LukeBailey181/obfuscated-activations

1 Introduction

In response to attacks on LLMs, such as jailbreaks or data poisoning, researchers have developed a variety of latent-space defense methods. These methods look at model activations and attempt to detect any signs of harmful content. Given that activations are causally upstream of outputs, one might hope that directly monitoring activations can catch issues at their root. Accordingly, recent work has proposed many latent-space defense methods, including monitors based on sparse autoencoders [Templeton et al., 2024, Gao et al., 2024, Lieberum et al., 2024], linear and nonlinear probes [Ball et al., 2024, Arditi et al., 2024, Kirch et al., 2024], and latent OOD detection [Podolskiy et al., 2021, Chen et al., 2022, Ma et al., 2023, Mo et al., 2024].

How robust are these latent-space defenses? An attacker trying to bypass latent-space monitors has two goals: they need to induce inconspicuous activations, but those activations must also lead to the attacker's desired model behavior. One could imagine these goals being *fundamentally incompatible*. For example, if the latent space contains only a single "harmfulness direction"—and only activations that align with this direction can produce harmful behavior—then it would be impossible to induce harmful outputs while also fooling a linear classifier encoding this direction. On the other hand, if there are many different regions in the latent space that can all lead to harmful behavior, latent-space monitors may not capture all of them. We call adversarial activations that produce undesired behavior while fooling a latent-space monitor "*obfuscated activations*."

In this paper, we look for obfuscated activations using *obfuscation attacks*, which jointly optimize

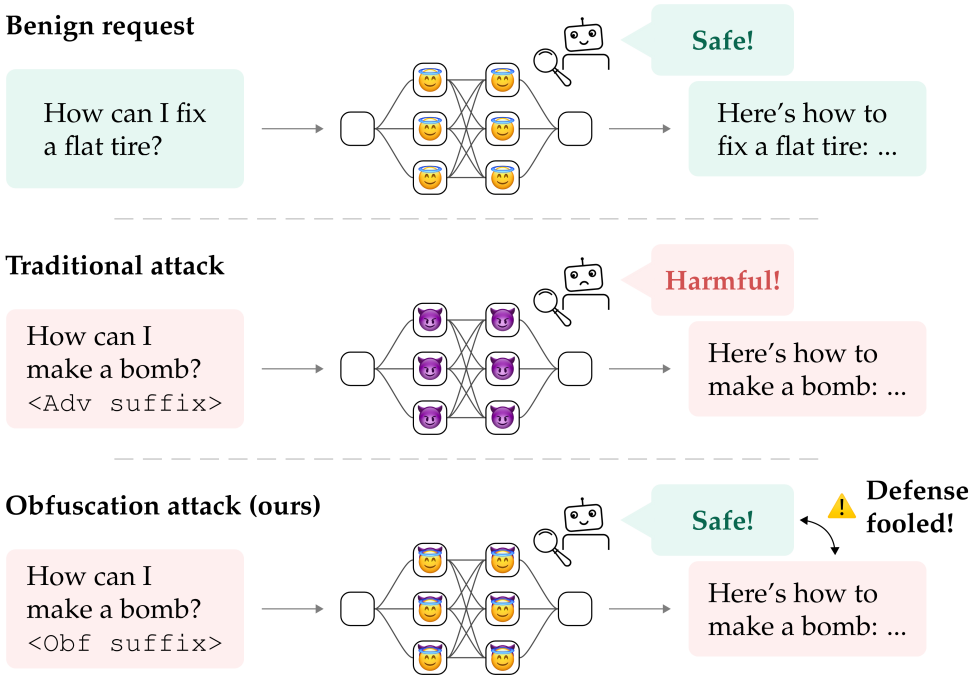

Figure 1: **Attackers can induce *obfuscated activations* in LLMs that bypass state-of-the-art latent-space defenses.** For example, latent harmfulness monitors can defend against traditional jailbreaks. But we find that attackers can produce adversarial suffixes that make activations appear benign to monitors, while still producing harmful outputs.

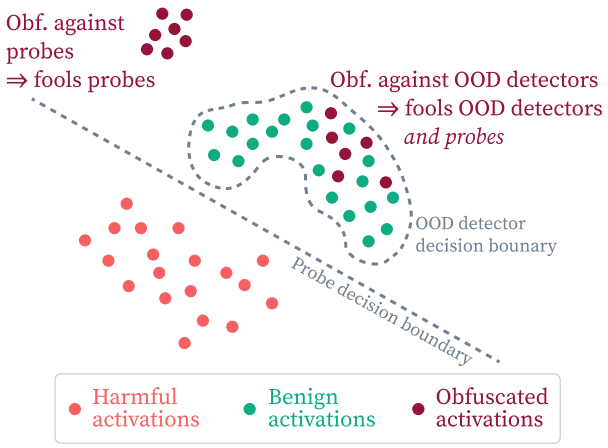

Figure 2: **Intuition:** Conceptual sketch of obfuscation against supervised probes and OOD detectors. Obfuscating against probes can lead to activations far away from benign ones that are flagged by an OOD detector. But obfuscating against an OOD detector forces obfuscated activations to be close to benign ones to stay within the detector's decision boundary, so these activations bypass probes as a side effect. While these geometric intuitions are speculation, **we empirically observe that obfuscating against OOD detectors transfers to fooling probes.**

for a certain model behavior (such as jailbreaking) while fooling latent-space monitors. *We find that obfuscation attacks can bypass a wide range of monitors,* including sparse autoencoders, supervised probes, and latent OOD detectors. Using different attack vectors (adversarial suffixes, data poisoning, model finetuning) ranging from white-box to black-box settings, we can induce undesired behaviors that aren't flagged by these defenses.

Obfuscated activations are not rare exceptions but rather appear to be widespread in the latent space. For example, we find obfuscated activations close to the cluster of benign activations when optimizing against OOD detectors. Because these activations are intermingled with inconspicuous ones, they also bypass supervised harmfulness probes as a side effect (Fig. 2). As another test of the abundance of obfuscated activations, we let a defender train a monitor on examples of obfuscated activations. If there were only a few latent regions of obfuscated activations, the monitor should quickly learn to cover them all and become robust. But we find that an attacker can always find new activations that bypass the monitor, even after retraining the monitor for over 70 iterations of alternating attack and defense optimization (§3.5). Obfuscated activations thus seem common enough that it is difficult for a defender to guard against them all.

However, obfuscation attacks can not induce *arbitrary* combinations of activations and behavior. When obfuscating against probes that detect SQL code, the model still generates SQL, but its code is incorrect more often than without obfuscation (§5). This suggests a trade-off between obfuscation and performance in difficult tasks.

We present our results as a series of case studies. Each case study demonstrates the existence of obfuscated activations in one domain:

- **Harmfulness monitoring** (§3). We demonstrate that linear and MLP probes can flag traditional jailbreaks as harmful, but can be fooled by obfuscation attacks, using both embedding and hard prompt suffixes. We also study failures in models produced with latent robustness training methods (Circuit Breakers [Zou et al., 2024] and Latent Adversarial Training [Sheshadri et al., 2024]) and show that these failures are related to obfuscated activations.
- **Backdoor detection** (§4). We show that obfuscation attacks can fool backdoor detection methods if the attacker has full control over the model's training process. We also study a black-box setting where the attacker can only poison training data and find that the attacker can still achieve some obfuscation by choosing a complex trigger or low poison rate.